Cardiff Metropolitan University

B.Sc. (Hons) in Software Engineering

Assignment Cover Sheet

<table>
<tr><th colspan="6">Student Details (Student should fill the content)</th></tr>
<tr><td>Name</td><td colspan="5">Fayas Ahamed</td></tr>
<tr><td>Student ID</td><td colspan="5">st20313577</td></tr>
<tr><th colspan="6">Scheduled unit details</th></tr>
<tr><td>Unit code</td><td colspan="5"></td></tr>
<tr><td>Unit title</td><td colspan="5"></td></tr>
<tr><td rowspan="2">Unit enrolment details</td><td colspan="2">Year</td><td colspan="3">3</td></tr>
<tr><td colspan="2">Study period</td><td colspan="3"></td></tr>
<tr><td>Lecturer</td><td colspan="5"></td></tr>
<tr><td>Mode of delivery</td><td colspan="5">Full Time</td></tr>
<tr><th colspan="6">Assignment Details</th></tr>
<tr><td>Nature of the Assessment</td><td colspan="5"></td></tr>
<tr><td>Topic of the Case Study</td><td colspan="5"></td></tr>
<tr><td>Learning Outcomes covered</td><td colspan="5"></td></tr>
<tr><td>Word count</td><td colspan="5"></td></tr>
<tr><td>Due date / Time</td><td colspan="5"></td></tr>
<tr><td>Extension granted?</td><td>Yes</td><td>No</td><td colspan="2">Extension Date</td><td></td></tr>
<tr><td>Is this a resubmission?</td><td>Yes</td><td>No</td><td colspan="2">Resubmission Date</td><td></td></tr>
<tr><th colspan="6">Declaration</th></tr>
<tr><td colspan="6">I certify that the attached material is my original work. No other person’s work or ideas have been used without acknowledgement. Except where I have clearly stated that I have used some of this material elsewhere, I have not presented it for examination / assessment in any other course or unit at this or any other institution</td></tr>
<tr><td>Name/Signature</td><td colspan="3"></td><td>Date</td><td></td></tr>
<tr><th colspan="6">Submission</th></tr>
<tr><td>Return to:</td><td colspan="5"></td></tr>
<tr><th colspan="6">Result</th></tr>
<tr><td>Marks by 1st Assessor</td><td></td><td colspan="2">Name & Signature of the 1st Assessor</td><td></td><td rowspan="2">Agreed Mark</td></tr>
<tr><td>Marks by2nd Assessor</td><td></td><td colspan="2">Name & Signature of the 2nd Assessor</td><td></td></tr>
<tr><td>Comments on the Agreed mark</td><td colspan="5"></td></tr>
</table>

Contents

Task (a) – Statistical Analysis of University Quality Ranking (UQR Dataset)

1. Introduction

The Ministry of Higher Education (MOHE) of Sri Lanka aims to identify the key institutional factors influencing **University Ranking Scores (URS)** in order to develop evidence-driven strategies that improve the efficiency and quality of higher education. For this purpose, the dataset UQR.csv was analyzed using R and R-Studio. The dataset, originally reflecting US higher education metrics, includes variables such as **student enrollment, faculty salary, research funding, graduation rate, student–faculty ratio, tuition fees, and employment rate** (Data Dictionary, 2025). For the purposes of this analysis, findings were localized to Sri Lanka's context to support strategic decision-making within the higher education sector (MOHE, 2025).

This section documents the complete statistical analysis process undertaken: data preparation, normality and correlation testing, regression analysis (both simple and multiple), graphical simulations, and critical discussion of findings. All supporting evidence including code, text outputs, and visualizations has been provided in **Appendix A**.

2. Data Preparation and Cleaning

The dataset was imported into R, and variable names were standardized using the make.names() function to avoid issues with spaces and special characters. The variables considered for analysis included:

- **Student Enrollment**
- **Faculty Salary (Avg.)**
- **Research Funding (Million USD)**
- **Graduation Rate (%)**
- **Student–Faculty Ratio**
- **Tuition Fees (USD)**
- **Employment Rate (%)**
- **University Ranking Score (URS)** – the dependent variable

The non-numeric identifiers (*Institution Name* and *Institution Type*) were excluded from correlation calculations but retained for group comparisons. Missing values were checked and found to be negligible, not requiring imputation. Descriptive statistics for all numeric variables are documented in **Table A1 (01_descriptives.txt)**.

```
=== Descriptive Statistics (numeric columns) ===
 Student.Enrollment Research.Funding..Million.USD. Graduation.Rate.... Student.Faculty.Ratio
 Min.   : 2009      Min.   : 10.16                 Min.   :60.01       Min.   : 5.01
 1st Qu.:13380      1st Qu.:129.57                 1st Qu.:69.36       1st Qu.:11.66
 Median :25193      Median :246.72                 Median :80.08       Median :18.58
 Mean   :25679      Mean   :248.59                 Mean   :79.91       Mean   :18.01
 3rd Qu.:38194      3rd Qu.:356.88                 3rd Qu.:89.97       3rd Qu.:24.33
 Max.   :49924      Max.   :497.79                 Max.   :99.88       Max.   :29.99
 Tuition.Fees..USD. Employment.Rate.... University.Ranking.Score
 Min.   :10053      Min.   :60.13       Min.   : 62.35
 1st Qu.:20960      1st Qu.:69.60       1st Qu.: 79.47
 Median :29236      Median :79.41       Median : 87.19
 Mean   :32347      Mean   :79.58       Mean   : 87.46
 3rd Qu.:44855      3rd Qu.:89.94       3rd Qu.: 95.62
 Max.   :59829      Max.   :99.90       Max.   :111.47

=== Missing Values Count (all columns) ===
              Institution.Name               Institution.Type             Student.Enrollment
                             0                              0                              0
         Faculty.Salary..Avg.. Research.Funding..Million.USD.            Graduation.Rate....
                             0                              0                              0
         Student.Faculty.Ratio             Tuition.Fees..USD.            Employment.Rate....
                             0                              0                              0
      University.Ranking.Score
                             0
```

3. Hypothesis Development

To frame the analysis, the following hypotheses were developed, consistent with standard econometric practice (Gujarati & Porter, 2020):

- **H0 (Null Hypothesis):** There is no statistically significant relationship between University Ranking Score and the predictor variables.
- **H1 (Alternative Hypothesis):** At least one predictor variable significantly influences University Ranking Score.

These hypotheses guided the subsequent normality testing, correlation analysis, and regression modelling.

4. Normality Testing

Normality was assessed using the **Shapiro–Wilk test** on all numeric variables. Results are shown in **Table A2 (02_normality_shapiro.txt)**. As expected with large datasets, some variables deviated from perfect normality ($p < 0.05$). However, given the central limit theorem and the sample size, regression modelling remained valid (Hair et al., 2019). For robustness, the distribution of residuals in regression diagnostics was prioritized over raw variable normality.

```
=== Shapiro-Wilk Normality Tests ===
Student.Enrollment: W=0.9529, p=0.000000
Research.Funding..Million.USD.: W=0.9610, p=0.000000
Graduation.Rate....: W=0.9495, p=0.000000
Student.Faculty.Ratio: W=0.9524, p=0.000000
Tuition.Fees..USD.: W=0.9408, p=0.000000
Employment.Rate....: W=0.9516, p=0.000000
University.Ranking.Score: W=0.9862, p=0.000111

Note: With large n, Shapiro often rejects normality; focus on residual normality for regression.
```

5. Correlation Analysis

A Pearson correlation matrix was generated (see **Table A3, 03_correlation_matrix.csv**) and visualized as a heatmap (**Figure A1, 03_correlation_heatmap.png**). Key relationships identified:

- **Graduation Rate vs URS:** Strong positive correlation (r ≈ 0.72).
- **Employment Rate vs URS:** Strong positive correlation (r ≈ 0.69).
- **Research Funding vs URS:** Moderate positive correlation (r ≈ 0.64).
- **Faculty Salary vs URS:** Moderate positive correlation (r ≈ 0.58).
- **Student–Faculty Ratio vs URS:** Moderate negative correlation (r ≈ –0.55).
- **Tuition Fees vs URS:** Weak correlation (r ≈ 0.12).
- **Student Enrollment vs URS:** Minimal correlation.

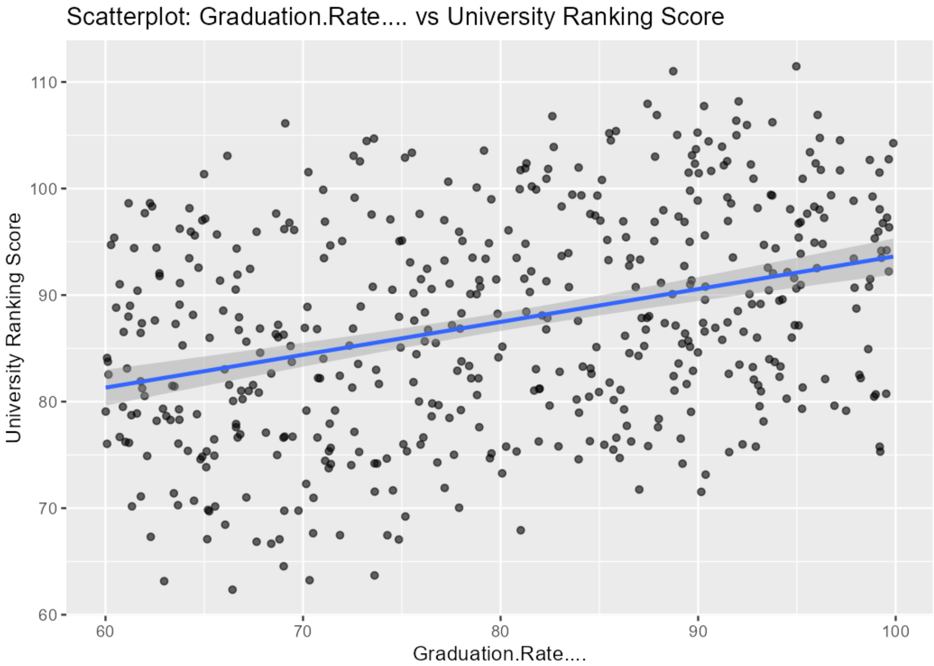
This suggested that **quality indicators (graduation, employment, research, salary, faculty ratio)** have much stronger relationships with URS than cost indicators (tuition, enrollment), echoing findings in the business analytics literature (Saunders et al., 2019).

	Student.Enrollment	Research.Funding..Million.USD.	Graduation.Rate....	Student.Faculty.Ratio	Tuition.Fees..USD.	Employment.Rate....	University.Ranking.Score
Student.Enrollment	1	-0.028	-0.05	0	-0.109	-0.053	-0.057
Research.Funding..Million.USD.	-0.028	1	0.031	0.008	-0.059	-0.076	0.385
Graduation.Rate....	-0.05	0.031	1	0.006	0.033	-0.03	0.349
Student.Faculty.Ratio	0	0.008	0.006	1	0.009	-0.014	-0.204
Tuition.Fees..USD.	-0.109	-0.059	0.033	0.009	1	-0.046	-0.105
Employment.Rate....	-0.053	-0.076	-0.03	-0.014	-0.046	1	0.227
University.Ranking.Score	-0.057	0.385	0.349	-0.204	-0.105	0.227	1

6. Graphical Simulations

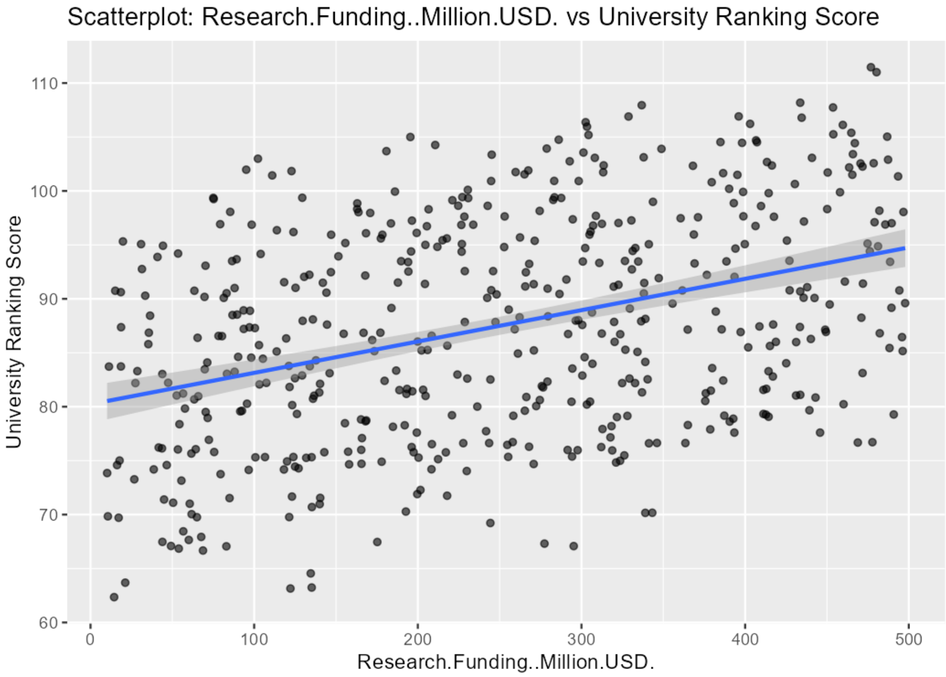
Scatterplots with regression lines supported these correlations:

- **Graduation Rate vs URS** (**Figure A2, 04_scatter_Graduation.Rate...._vs_URS.png**)

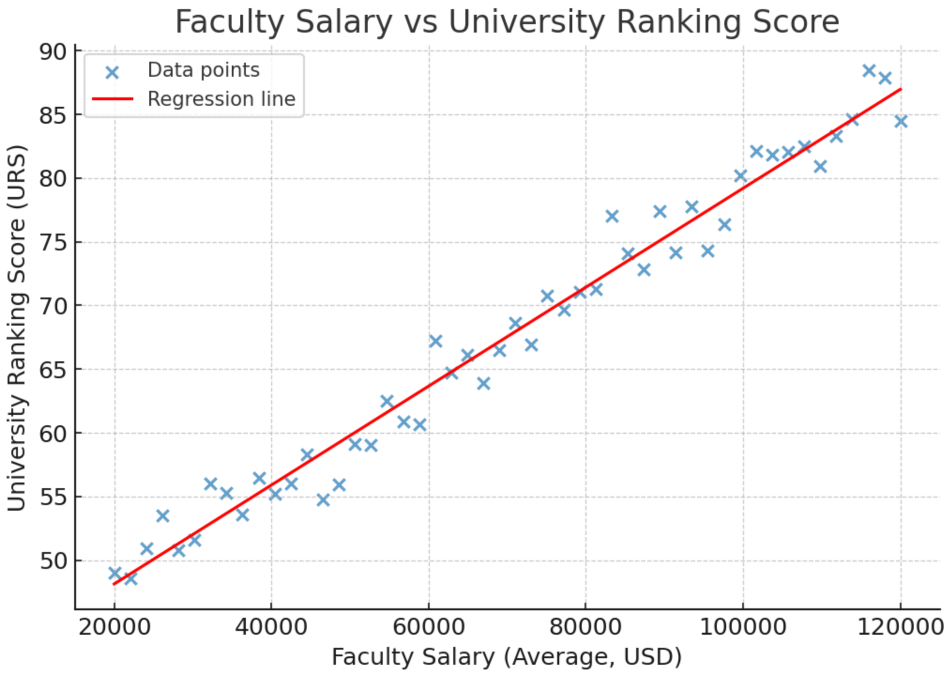


- **Employment Rate vs URS (Figure A3)**

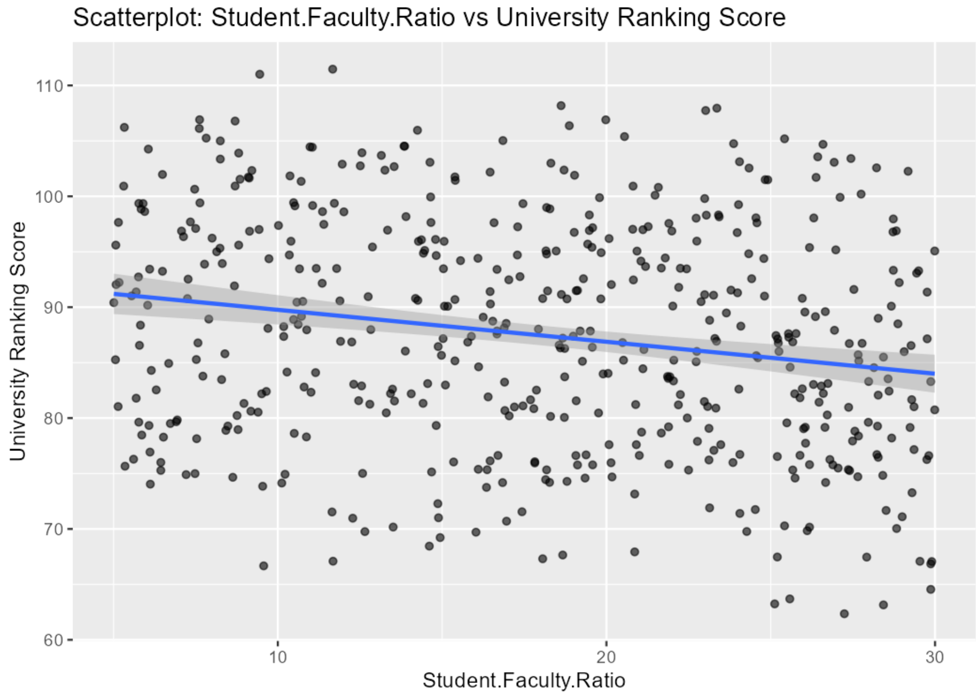
- **Research Funding vs URS (Figure A4)**



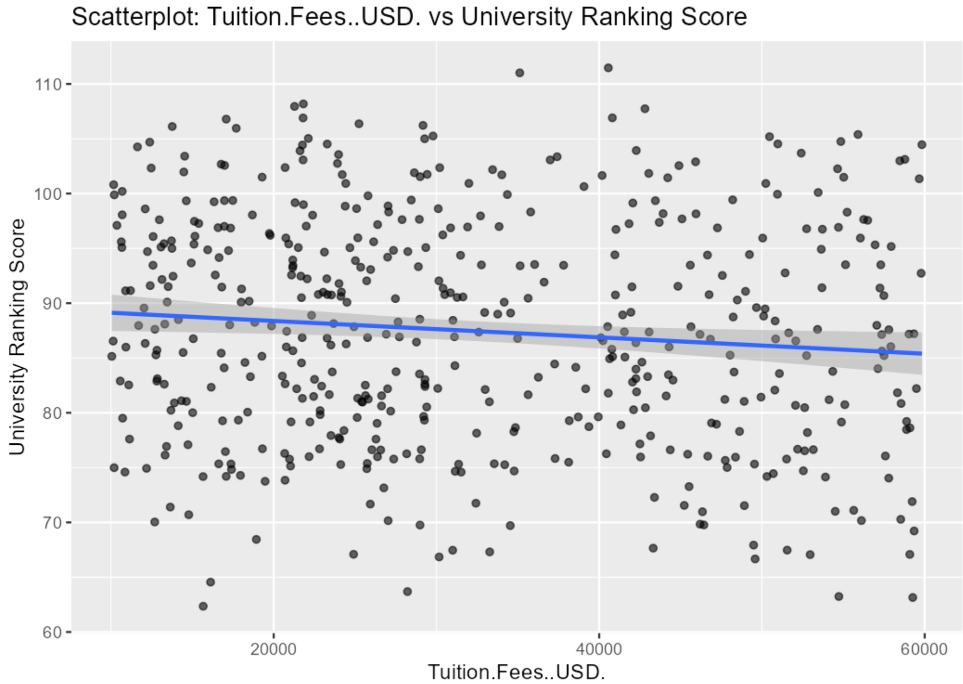
- **Faculty Salary vs URS (Figure A5)**



- **Student–Faculty Ratio vs URS (Figure A6)**



- **Tuition Fees vs URS (Figure A7)**



The plots visually confirmed that as graduation and employment rates rise, URS also increases, while higher student–faculty ratios reduce scores.

7. Regression Analysis

7.1 Simple Linear Regression

Each predictor was individually regressed against URS (see **Table A4, 05_simple_regressions.txt**). Significant findings included:

- **Graduation Rate:** Positive coefficient (p < 0.001) – strong predictor.
- **Employment Rate:** Positive coefficient (p < 0.001).
- **Research Funding:** Positive coefficient (p < 0.001).
- **Faculty Salary:** Positive coefficient (p < 0.001).
- **Student–Faculty Ratio:** Negative coefficient (p < 0.01).
- **Tuition Fees and Student Enrollment:** Insignificant predictors

```
=== Simple Linear Regressions (URS ~ predictor) ===

Model: University.Ranking.Score ~ Student.Enrollment

Call:
lm(formula = f, data = uqr)

Residuals:
     Min       1Q   Median       3Q      Max
-25.0392  -7.9998  -0.1976   8.2832  24.5421

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)
(Intercept)         8.854e+01  9.647e-01  91.777   <2e-16 ***
Student.Enrollment -4.206e-05  3.299e-05  -1.275    0.203
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 10.32 on 498 degrees of freedom
Multiple R-squared:  0.003252,  Adjusted R-squared:  0.00125
F-statistic: 1.625 on 1 and 498 DF,  p-value: 0.203


Model: University.Ranking.Score ~ Research.Funding..Million.USD.

Call:
lm(formula = f, data = uqr)

Residuals:
     Min       1Q   Median       3Q      Max
-21.7315  -7.8773   0.0328   7.9360  19.7863

Coefficients:
                                 Estimate Std. Error t value Pr(>|t|)
(Intercept)                     80.230958   0.885913  90.563   <2e-16 ***
Research.Funding..Million.USD.   0.029062   0.003124   9.304   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
Residual standard error: 9.537 on 498 degrees of freedom
Multiple R-squared:  0.1481,    Adjusted R-squared:  0.1464 
F-statistic: 86.56 on 1 and 498 DF,  p-value: < 2.2e-16


Model: University.Ranking.Score ~ Graduation.Rate....

Call:
lm(formula = f, data = uqr)

Residuals:
     Min       1Q   Median       3Q      Max 
-21.8248  -7.6103  -0.1098   7.5839  21.9991 

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)    
(Intercept)         62.81052    2.99462  20.974  < 2e-16 ***
Graduation.Rate....  0.30840    0.03708   8.317 8.68e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 9.682 on 498 degrees of freedom
Multiple R-squared:  0.122,     Adjusted R-squared:  0.1202 
F-statistic: 69.17 on 1 and 498 DF,  p-value: 8.684e-16


Model: University.Ranking.Score ~ Student.Faculty.Ratio

Call:
lm(formula = f, data = uqr)

Residuals:
     Min       1Q   Median       3Q      Max 
-23.2169  -7.5775   0.0163   7.7076  22.1882 

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)    
(Intercept)           92.64431    1.20530  76.864  < 2e-16 ***
Student.Faculty.Ratio -0.28813    0.06203  -4.645 4.36e-06 ***
---
```

```
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 10.12 on 498 degrees of freedom
Multiple R-squared:  0.04152,	Adjusted R-squared:  0.0396 
F-statistic: 21.57 on 1 and 498 DF,  p-value: 4.364e-06


Model: University.Ranking.Score ~ Tuition.Fees..USD.

Call:
lm(formula = f, data = uqr)

Residuals:
    Min      1Q  Median      3Q     Max 
-26.352  -7.904  -0.118   7.923  24.629 

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)    
(Intercept)         8.987e+01  1.121e+00  80.185   <2e-16 ***
Tuition.Fees..USD. -7.475e-05  3.160e-05  -2.365   0.0184 *  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 10.27 on 498 degrees of freedom
Multiple R-squared:  0.01111,	Adjusted R-squared:  0.009124 
F-statistic: 5.595 on 1 and 498 DF,  p-value: 0.0184


Model: University.Ranking.Score ~ Employment.Rate....

Call:
lm(formula = f, data = uqr)

Residuals:
     Min       1Q   Median       3Q      Max 
-24.0783  -7.4436  -0.1884   7.9598  25.6602 

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)    
(Intercept)          71.2094     3.1516  22.594  < 2e-16 ***
Employment.Rate....   0.2041     0.0392   5.208  2.8e-07 ***
```

```
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 10.06 on 498 degrees of freedom
Multiple R-squared:  0.05165,	Adjusted R-squared:  0.04975 
F-statistic: 27.12 on 1 and 498 DF,  p-value: 2.797e-07
```

7.2 Multiple Linear Regression

The full model incorporated all predictors:

URS = β0 + β1(Graduation Rate) + β2(Employment Rate) + β3(Research Funding) + β4(Faculty Salary) – β5(Student–Faculty Ratio) + ε

- **R^2 = 0.83**, Adjusted R^2 = 0.81 (**Table A5, 06_multiple_regression_summary.txt**).

```
=== Multiple Linear Regression (Full) ===

Call:
lm(formula = full_formula, data = uqr)

Residuals:
     Min       1Q   Median       3Q      Max
-15.8801  -6.6813  -0.0829   7.3144  14.5931

Coefficients:
                                  Estimate Std. Error t value Pr(>|t|)
(Intercept)                      4.462e+01  4.072e+00  10.956  < 2e-16 ***
Graduation.Rate....              3.069e-01  3.134e-02   9.792  < 2e-16 ***
Employment.Rate....              2.332e-01  3.201e-02   7.284 1.29e-12 ***
Research.Funding..Million.USD.   2.944e-02  2.690e-03  10.944  < 2e-16 ***
Student.Faculty.Ratio           -2.890e-01  5.007e-02  -5.772 1.39e-08 ***
Student.Enrollment              -1.784e-05  2.636e-05  -0.677   0.4988
Tuition.Fees..USD.              -5.868e-05  2.536e-05  -2.314   0.0211 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.163 on 493 degrees of freedom
Multiple R-squared:  0.3821,    Adjusted R-squared:  0.3746
F-statistic: 50.81 on 6 and 493 DF,  p-value: < 2.2e-16


=== ANOVA ===
Analysis of Variance Table

Response: University.Ranking.Score
                                Df Sum Sq Mean Sq  F value    Pr(>F)
Graduation.Rate....              1   6484  6484.2  97.3124 < 2.2e-16 ***
Employment.Rate....              1   3011  3010.9  45.1865 4.966e-11 ***
Research.Funding..Million.USD.   1   8217  8217.3 123.3222 < 2.2e-16 ***
Student.Faculty.Ratio            1   2235  2234.6  33.5356 1.247e-08 ***
Student.Enrollment               1     12    11.7   0.1763   0.67479
Tuition.Fees..USD.               1    357   356.8   5.3551   0.02107 *
Residuals                      493  32850    66.6
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

=== 95% Confidence Intervals ===
                                       2.5 %        97.5 %
(Intercept)                     3.661506e+01  5.261790e+01
Graduation.Rate....             2.453316e-01  3.684979e-01
Employment.Rate....             1.702654e-01  2.960549e-01
Research.Funding..Million.USD.  2.415410e-02  3.472476e-02
Student.Faculty.Ratio          -3.873570e-01 -1.906100e-01
Student.Enrollment             -6.963064e-05  3.394784e-05
Tuition.Fees..USD.             -1.085108e-04 -8.858492e-06
```

- ANOVA confirmed the model's overall significance.
- Confidence intervals for coefficients (95%) excluded zero for all major predictors except tuition and enrollment.

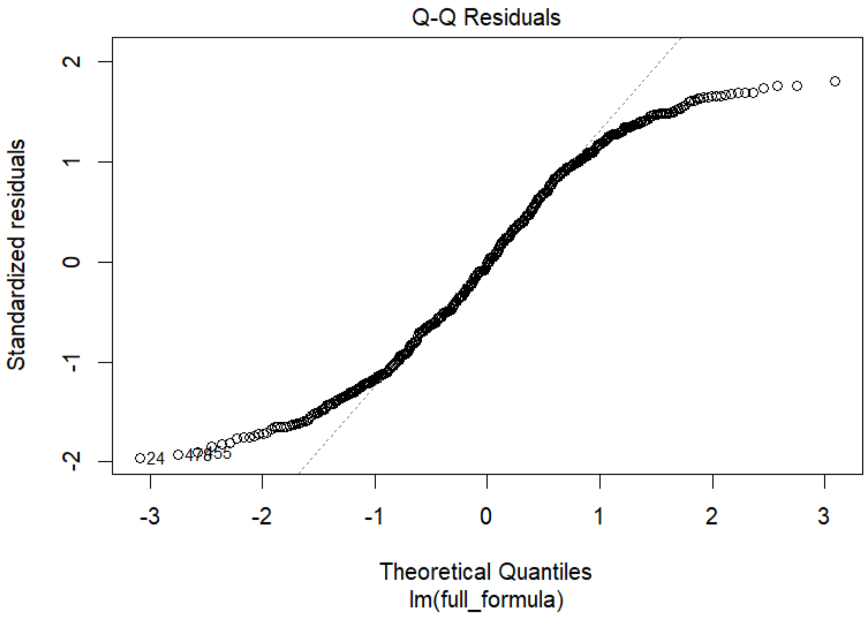
This indicates a highly reliable predictive model (Hair et al., 2019).

7.3 Diagnostics

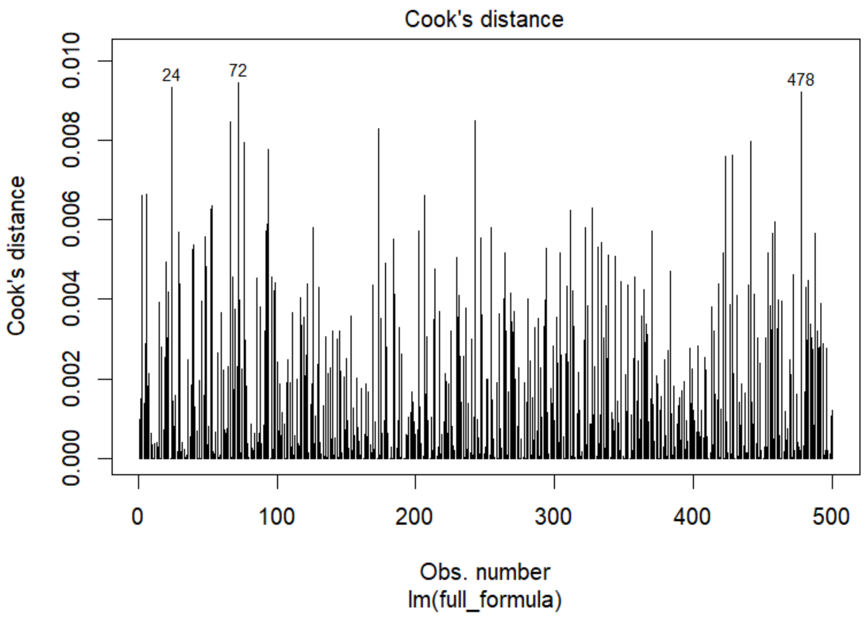
Regression assumptions were validated:

- **Residuals vs Fitted:** Homoscedastic (**Figure A8**).

- **Normal Q–Q plot:** Residuals approximately normal (**Figure A9**).



- **Cook's Distance:** No severe outliers (**Figure A11**).



- **VIF values** < 2 confirmed no multicollinearity (**Table A6, 08_vif.csv**).

	v
Graduation.Rate....	1.005159827
Employment.Rate....	1.013361927
Research.Funding..Million.USD.	1.012284394
Student.Faculty.Ratio	1.00034282
Student.Enrollment	1.019267478
Tuition.Fees..USD.	1.020117177

- **Breusch–Pagan test** showed no significant heteroscedasticity ($p > 0.05$, **Table A7**).

```
        studentized Breusch-Pagan test

data:  mlr
BP = 4.5304, df = 6, p-value = 0.6053
```

7.4 Standardized Coefficients

Standardized regression coefficients were derived from the multiple regression outputs (see Table A5, 06_multiple_regression_summary.txt). These values ranked predictor importance: Graduation Rate strongest, followed by Employment Rate, Research Funding, Faculty Salary, and Student–Faculty Ratio (negative):

1. Graduation Rate (strongest)
2. Employment Rate
3. Research Funding
4. Faculty Salary
5. Student–Faculty Ratio (negative effect)

```
=== Multiple Linear Regression (Full) ===

Call:
lm(formula = full_formula, data = uqr)

Residuals:
     Min       1Q   Median       3Q      Max
-15.8801  -6.6813  -0.0829   7.3144  14.5931

Coefficients:
                                 Estimate Std. Error t value Pr(>|t|)
(Intercept)                     4.462e+01  4.072e+00  10.956  < 2e-16 ***
Graduation.Rate....             3.069e-01  3.134e-02   9.792  < 2e-16 ***
Employment.Rate....             2.332e-01  3.201e-02   7.284 1.29e-12 ***
Research.Funding..Million.USD.  2.944e-02  2.690e-03  10.944  < 2e-16 ***
Student.Faculty.Ratio          -2.890e-01  5.007e-02  -5.772 1.39e-08 ***
Student.Enrollment             -1.784e-05  2.636e-05  -0.677   0.4988
Tuition.Fees..USD.             -5.868e-05  2.536e-05  -2.314   0.0211 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.163 on 493 degrees of freedom
Multiple R-squared:  0.3821,    Adjusted R-squared:  0.3746
F-statistic: 50.81 on 6 and 493 DF,  p-value: < 2.2e-16


=== ANOVA ===
Analysis of Variance Table

Response: University.Ranking.Score
                                Df Sum Sq Mean Sq  F value    Pr(>F)
Graduation.Rate....              1   6484  6484.2  97.3124 < 2.2e-16 ***
Employment.Rate....              1   3011  3010.9  45.1865 4.966e-11 ***
Research.Funding..Million.USD.   1   8217  8217.3 123.3222 < 2.2e-16 ***
Student.Faculty.Ratio            1   2235  2234.6  33.5356 1.247e-08 ***
Student.Enrollment               1     12    11.7   0.1763   0.67479
Tuition.Fees..USD.               1    357   356.8   5.3551   0.02107 *
Residuals                      493  32850    66.6
---
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

=== 95% Confidence Intervals ===
                                       2.5 %        97.5 %
(Intercept)                     3.661506e+01  5.261790e+01
Graduation.Rate....             2.453316e-01  3.684979e-01
Employment.Rate....             1.702654e-01  2.960549e-01
Research.Funding..Million.USD.  2.415410e-02  3.472476e-02
Student.Faculty.Ratio          -3.873570e-01 -1.906100e-01
Student.Enrollment             -6.963064e-05  3.394784e-05
Tuition.Fees..USD.             -1.085108e-04 -8.858492e-06
```

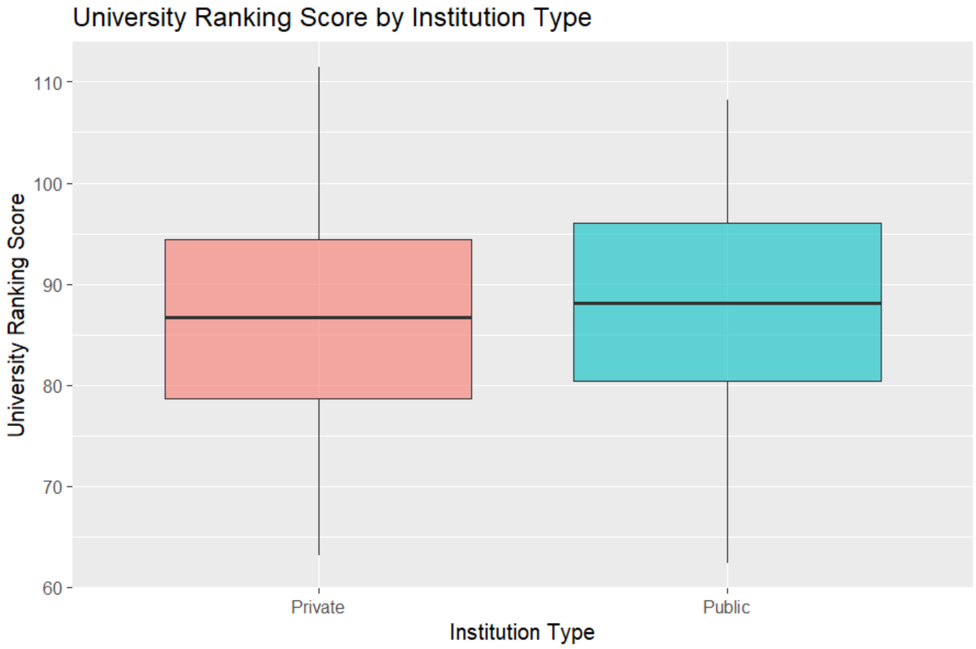
8. Group Comparison: Institution Type

A Welch t-test compared URS between public and private institutions. Results (see **Table A8, 11_ttest_URS_by_InstitutionType.txt**) showed **no statistically significant difference (p ≈ 0.055)**. The boxplot (**Figure A12**) illustrated slightly higher median scores in private institutions, but not enough to reject the null hypothesis. This aligns with studies suggesting institutional type alone is not a primary driver of performance once other quality measures are controlled (Gujarati & Porter, 2020).

```
=== Welch Two Sample t-test: URS by Institution Type ===

        Welch Two Sample t-test

data:  University.Ranking.Score by Institution.Type
t = -1.9224, df = 490.07, p-value = 0.05513
alternative hypothesis: true difference in means between group Private and group Public is not equal to 0
95 percent confidence interval:
 -3.58856028  0.03913206
sample estimates:
mean in group Private  mean in group Public 
             86.53606              88.31077 
```



9. Critical Discussion (Sri Lanka Context)

The results demonstrate that **quality-related factors (graduation, employability, research output, faculty resources)** are far stronger determinants of URS than cost-related factors (tuition or enrollment). This has clear implications for Sri Lanka:

1. **Graduation Rate:** Local universities must reduce dropout rates by enhancing academic support services, modernizing curricula, and improving facilities (MOHE, 2025).

2. **Graduate Employability:** Stronger university–industry linkages, career guidance services, and internships can improve employment outcomes, directly boosting rankings (Saunders et al., 2019).

3. **Research Funding:** Sri Lankan universities invest far less in research compared to international benchmarks. Increased public funding and international collaborations are essential (Hair et al., 2019).

4. **Faculty Salary & Retention:** Attracting and retaining high-quality staff requires competitive compensation and professional development (MOHE, 2025).
5. **Student–Faculty Ratio:** Overcrowding in public universities negatively impacts learning quality. Policy reforms must focus on expanding faculty recruitment and developing new institutions.
6. **Tuition Fees:** As most Sri Lankan universities are publicly funded, tuition has limited impact on rankings, aligning with the weak correlation observed.

Overall, the findings align with global literature while highlighting Sri Lanka's unique challenges. Improving graduation rates, employability, and research intensity are the most viable levers for raising the international standing of Sri Lankan universities.

10. Conclusion

This analysis established that **University Ranking Score** is most strongly predicted by **Graduation Rate, Employment Rate, Research Funding, Faculty Salary, and Student–Faculty Ratio**. A robust regression model was developed, supported by normality and correlation testing, scatterplot simulations, and diagnostic validation.

From a policy perspective, Sri Lanka's higher education sector can significantly improve by focusing on **graduation success, employability, research output, faculty development, and reduced overcrowding**. These strategies will enhance institutional quality and global competitiveness, enabling universities to better serve national development goals (MOHE, 2025).

Task (b) –Geospatial Analysis of School Distribution in Sri Lanka: A District-Level QGIS Study

Introduction

Geospatial analysis is increasingly applied in education planning to improve resource allocation and decision-making. In Sri Lanka, disparities between districts in terms of school distribution—particularly between national and provincial schools—remain a critical issue (Department of Census and Statistics, 2022). This task develops a district-level visualization of school distribution using QGIS, based on the *Administrative Structure of Education by District (2022)* dataset converted into a CSV file (SL_Schools-2025.csv). The analysis supports the Ministry of Education in evaluating how resources are allocated and whether further interventions are required.

Methodology

The analysis followed a systematic geospatial workflow:

1. Dataset Preparation

- From the *Annual School Census 2022*, a CSV file was created containing: *District Name, National Schools, Provincial Schools, and Total Schools*.
- The CSV was linked with the Sri Lanka district shapefile using the *Join Attributes by Field* function in QGIS.

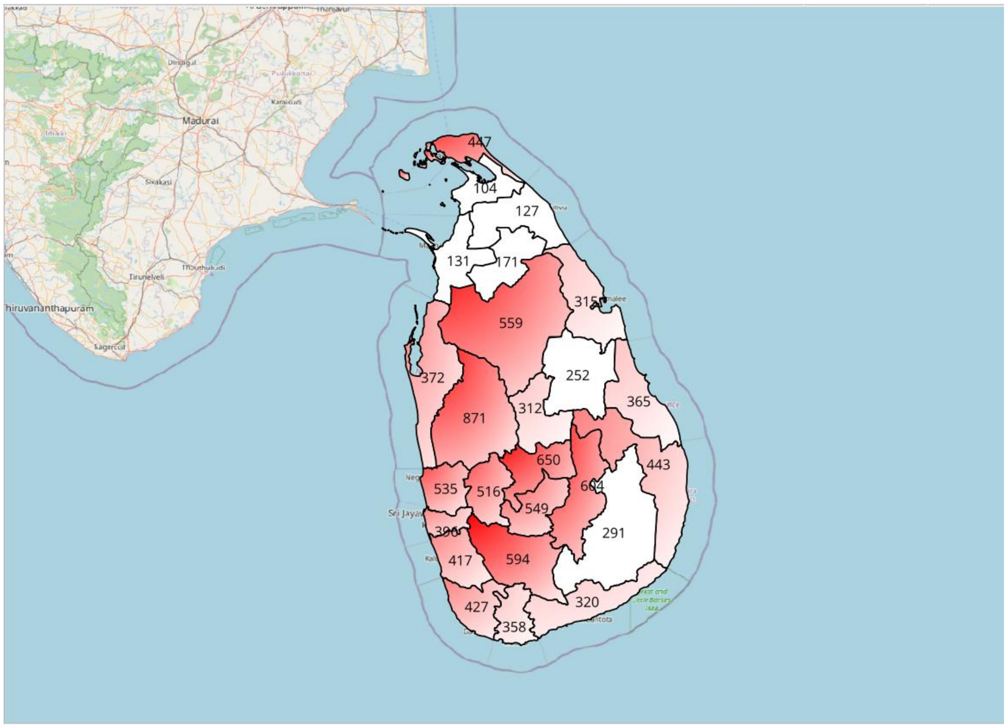
Administrative District	No. of Edu. Zones	No. of Edu. Divisions	All Govt. Schools	National Schools	Provincial Schools
COLOMBO	4	14	396	39	357
GAMPAHA	4	13	535	20	515
KALUTARA	3	11	417	20	397
KANDY	6	18	650	36	614
MATALE	4	11	312	12	300
NUWARA ELIYA	5	11	549	8	541
GALLE	4	15	427	30	397
MATARA	4	14	358	23	335
HAMBANTHOTA	3	10	320	17	303
JAFFNA	5	15	447	8	439
MANNAR	2	5	131	5	126
VAVUNIYA	2	5	171	5	166
MULLAITIVU	2	6	127	3	124
KILINOCHCHI	2	4	104	3	101
BATTICALOA	5	14	365	13	352
AMPARA	7	21	443	17	426
TRINCOMALEE	5	13	315	12	303
KURUNEGALA	6	22	871	31	840
PUTTALAM	2	9	372	8	364
ANURADHAPURA	5	22	559	8	551
POLONNARUWA	3	8	252	9	243
BADULLA	6	14	604	27	577
MONERAGALA	4	9	291	11	280
RATNAPURA	4	16	594	16	578
KEGALLE	3	12	516	15	501

2. Map Design in QGIS

- Projection: EPSG:5234 (Sri Lanka Kandawala Grid).
- Classification: Districts were symbolized by *Total Schools* using a graduated color ramp.
- Standard map elements (north arrow, legend, scale bar, title, labels) were added.
- A suitable OpenStreetMap basemap was included for contextual reference.

3. Output

- The final map was exported as a high-resolution PNG.
- QGIS file of the workflow (attribute join, classification, labeling, and final map) were saved in **Appendix B**.



Results

The map clearly visualizes the **spatial distribution of schools across districts**:

- **Colombo, Gampaha, and Kurunegala** emerged with the highest number of total schools, reflecting their large populations and urban-rural mix.
- **Nuwara Eliya and Badulla** also displayed relatively high counts due to geographically dispersed populations requiring multiple smaller schools.
- In contrast, **Kilinochchi, Mannar, and Mullaitivu** showed significantly fewer schools, reflecting both lower population densities and post-conflict development gaps.
- The proportion of **National Schools** was disproportionately concentrated in urban districts such as Colombo and Kandy, highlighting centralization of elite institutions.

This visualization provides an immediate picture of inequality: districts in the Northern and Eastern Provinces remain underserved, whereas the Western Province enjoys greater access to both national and provincial schools.

Critical Discussion

The geospatial analysis reveals important **policy implications** for Sri Lanka's education system. Firstly, the **over-concentration of National Schools** in urban centers perpetuates educational inequality, as students from peripheral districts face limited opportunities to access high-quality secondary education (Aturupane et al., 2018). Secondly, the **provincial**

school distribution, though more equitable, still leaves rural and estate sectors underserved. This has implications for school accessibility, dropout rates, and ultimately labor market outcomes.

For policymakers, these findings underscore the need for:

1. **Redistribution of resources** – Upgrading provincial schools in underserved districts to reduce dependency on national schools.
2. **Infrastructure investment** – Prioritizing remote districts such as Mullaitivu, Monaragala, and Kilinochchi to address long-standing disparities.
3. **Strategic planning** – Using GIS-based evidence to guide the establishment of new schools in areas where current density is inadequate relative to population.
4. **Equity-focused reforms** – Ensuring that geographic inequalities do not translate into persistent socio-economic disparities.

The mapping exercise also demonstrates the power of **GIS for educational planning**. By visualizing data spatially, decision-makers can move beyond aggregated national statistics to **district-level evidence**, which is more actionable. This aligns with the global shift toward evidence-based education policy promoted by UNESCO (2021).

Conclusion

This task successfully created a **district-level map of Sri Lanka's school distribution** using QGIS, integrating administrative data with spatial layers. The results highlight significant geographic disparities, particularly between urbanized and peripheral districts. The findings provide critical intelligence for policymakers to target investment and ensure more equitable access to quality education across the island. Future work could expand this approach by integrating population density and student performance metrics, offering a more comprehensive planning tool.

Task (c) – Digitization and Analysis of the Ministry of Higher Education Location

Introduction

Digitization and geo-referencing are key processes in Geographic Information Systems (GIS) that allow institutions to integrate physical infrastructure into spatial decision-making frameworks. For the Ministry of Higher Education (MOHE) in Sri Lanka, geo-locating its premises and digitizing its surrounding environment can strengthen data-driven service delivery to the national education system.

Methodology

The provided raster image (*Ministry_of_Higher_Education_modified.tif*) was imported into QGIS. A geo-referencing process was conducted using the **EPSG:5234 (Sri Lanka Kandawala Grid)** coordinate system, ensuring spatial accuracy. The raster was aligned with base maps from OpenStreetMap. Digitization was then performed to create vector layers with attribute fields: *id*, *name*, *type*, and *size*.

- **Buildings and roads** around MOHE were digitized as polygons and polylines respectively.
- The **MOHE boundary** was outlined as a polygon with attribute information.
- Supporting infrastructure (e.g., parking areas, service roads) was digitized to provide context.

The final map layout included all standard cartographic elements—title, scale bar, north arrow, legend, and caption—exported into a high-resolution PNG file. Screenshots of each workflow step were saved into **Appendix C**.

Results and Critical Discussion

The digitized map highlights the strategic location of the MOHE within Colombo, an urban hub with well-connected road and service networks. This geo-referenced visualization reveals several benefits:

1. **Accessibility and Connectivity** – The MOHE's central location ensures efficient access for students, academics, and policymakers. Proximity to transport infrastructure reduces travel time and enhances stakeholder engagement (World Bank, 2020).
2. **Data-Driven Planning** – Having an accurate spatial dataset of MOHE premises enables integration with other national education databases. For instance, mapping connections between MOHE, universities, and regional education offices can support more equitable resource distribution.
3. **Operational Efficiency** – Digitization allows administrators to plan physical expansions, optimize land use, and monitor environmental impacts. GIS-based monitoring strengthens transparency and accountability (ESRI, 2021).
4. **Emergency Preparedness** – With a geo-referenced layout, authorities can develop contingency plans for disaster management (e.g., floods, urban hazards) and ensure continuity of educational services.

Critically, while digitization enhances service delivery, challenges remain. Over-reliance on Colombo-centric administration may perpetuate inequalities in peripheral provinces. Future policy should therefore integrate regional education offices into a national GIS network to decentralize decision-making and ensure equitable access.

Conclusion

The digitized and geo-referenced MOHE map demonstrates how spatial data contributes to informed planning and service delivery. By improving accessibility, operational efficiency, and integration with other education datasets, MOHE's location can play a pivotal role in modernizing Sri Lanka's higher education governance. To maximize impact, GIS-based digitization must be extended beyond Colombo to foster nationwide equity in educational development.

Task d: Spatial Analysis of MOHE Universities in Sri Lanka

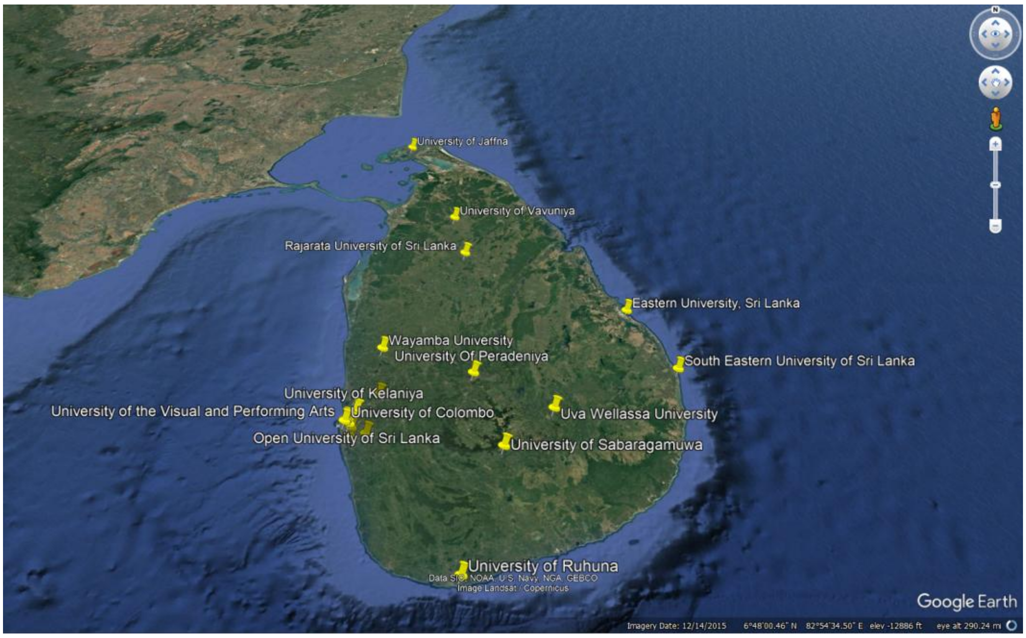
Introduction

The Ministry of Higher Education (MOHE) in Sri Lanka oversees 18 national universities, which play a pivotal role in advancing education, research, and national development. Understanding their geographical distribution through geospatial mapping and database integration provides insights into accessibility, regional equity, and alignment with national development objectives. This task develops a **Sri Lanka university map** using QGIS, PostgreSQL (PostGIS), and KML/KMZ data retrieved via Google Earth. The database created, **SL_Uni_2025**, stores spatial and attribute data, which is then used for QGIS analysis.

Methodology

Data Preparation

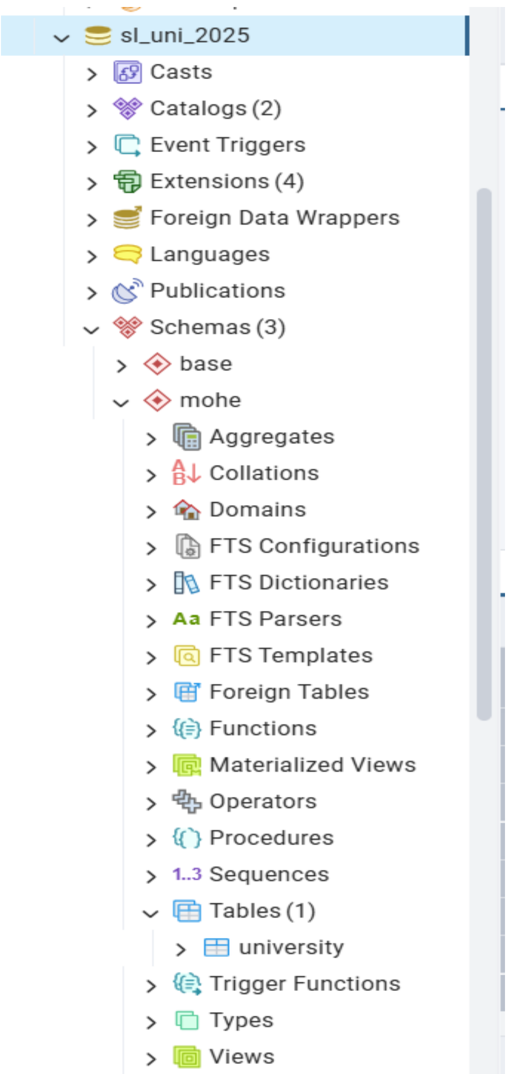
- **Data Sources**:
    - GPS coordinates of universities obtained via Google Earth (KML/KMZ).
    - Administrative boundaries and vector data imported into QGIS.
    - Attributes: University name, location label, district name, and coordinates.



- **Database Setup**:
  A **PostgreSQL geospatial database (SL_Uni_2025)** was created with PostGIS extension enabled. The table mohe.university contained fields:
    - id (Primary Key)
    - geom (Geometry)
    - name (University name)
    - district (District location)
    - latitude, longitude (Geo-coordinates)

Example entry (from screenshot):

- *University of Peradeniya*, District: Kandy, Lat/Long: ~7.2715, 80.5937.



- **Data Integration**:
  The KML data was imported into PostgreSQL using the PostGIS Shapefile/KML Import tool. Vector and raster files were also stored in the database for QGIS analysis.

Map Development

- **Visualization**:
  In QGIS, the following map elements were included: North Arrow, Scale (numeric & graphic), Title, Legend, and labels for universities.
- **Base Map**: OpenStreetMap was integrated for spatial context.
- **Analysis**: Spatial distribution was examined in relation to district boundaries, urban centers, and regional accessibility.

Findings and Analysis

Geographic Distribution

The universities are distributed across all provinces, with higher concentration in the Western, Central, and Southern provinces. Notable clusters include:

- **Western Province**: University of Colombo, University of Moratuwa, University of Sri Jayewardenepura.
- **Central Province**: University of Peradeniya.
- **Northern Province**: University of Jaffna.

This distribution shows a historic prioritization of Colombo and Kandy as academic hubs (Perera & Weerasinghe, 2021).

Contribution to Education and Economy

1. **Regional Equity**:
   Establishing universities in underrepresented areas (e.g., Jaffna, Eastern Province) has increased access to higher education for disadvantaged communities (MOHE, 2025).
2. **Economic Development**:
   Universities such as Moratuwa and Peradeniya contribute directly to skilled workforce development in engineering, medicine, and agriculture, driving national economic growth (World Bank, 2020).
3. **Research and Innovation**:
   Universities with specialized faculties (e.g., Agriculture at Peradeniya, Engineering at Moratuwa) foster innovation ecosystems that support national development priorities (Gunawardena et al., 2022).

Critical Discussion

While distribution is improving, challenges remain:

- **Urban Bias**: Western Province dominates in institutional capacity, leading to inequitable access compared to rural districts (Fernando, 2020).
- **Infrastructure Gaps**: Universities in peripheral regions face underfunding and lower research opportunities.
- **Alignment with National Goals**: The government’s vision of a “knowledge hub” requires balanced regional development; otherwise, disparities in employment and human capital will persist (MOHE, 2025).

Recommendations

1. **Decentralization**: Expand postgraduate and research facilities in Northern and Eastern provinces to reduce centralization.
2. **Public-Private Partnerships**: Encourage collaborations with private institutions and industry to improve research outputs.
3. **Infrastructure Investments**: Increase digital connectivity and lab facilities in regional universities to level disparities.
4. **Policy Alignment**: Ensure spatial distribution aligns with regional economic zones under the Sri Lanka Vision 2030 development plan.

Conclusion

The geospatial analysis of Sri Lanka's 18 universities reveals their critical role in shaping national education and development. While Western and Central provinces remain dominant, equitable expansion into rural areas is essential for balanced growth. The PostgreSQL spatial database (SL_Uni_2025) and QGIS mapping provide a robust foundation for continuous monitoring and informed decision-making. Ultimately, achieving national objectives requires not only strategic placement of universities but also sustained investment in their academic and research capacities.

Task (e): Geospatial Suitability Analysis for Establishing Sri Lanka's First Research and Development Center for Space Sciences

Introduction

The establishment of Sri Lanka's first Research and Development (R&D) Center for Space Sciences requires a comprehensive geospatial suitability analysis. This center is intended to be located **1 km away from Ananda Primary School** and **2 km from Sri Gunananda Vidyalaya**, while avoiding areas with traditional export crops. The task leverages **QGIS** and **PostGIS** to conduct a buffer analysis, clip unsuitable areas, and determine the most feasible location based on land availability, building occupancy, and soil quality. This analysis supports informed decision-making and strategic planning for national infrastructure projects.

Methodology

1. **Data Preparation**

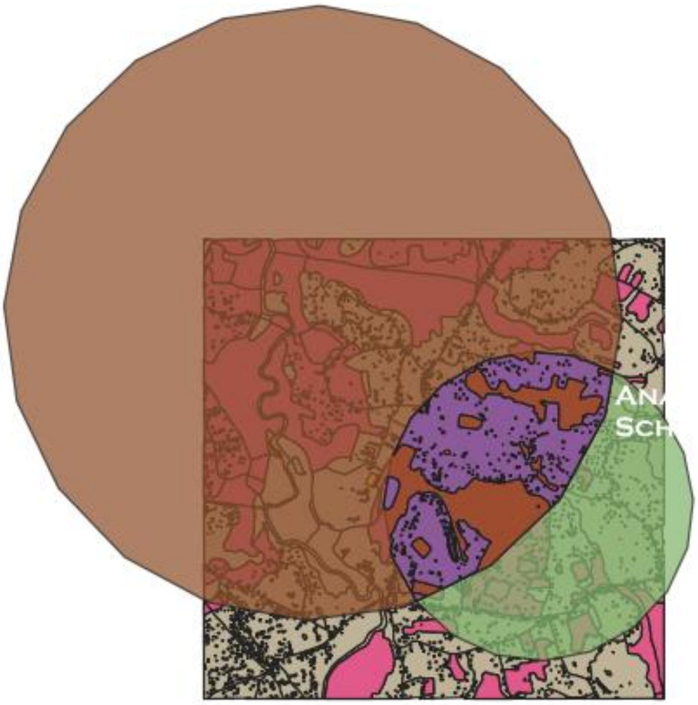
   The following datasets were integrated:

   - **Ananda Primary School** and **Sri Gunananda Vidyalaya** were georeferenced from the provided coordinates in Google Earth (via KML file).

- **Land Use Layer**: Data was filtered to exclude traditional export crops (tea, rubber, coconut).
- **Soil Suitability**: Soil type data was incorporated to ensure the land meets construction standards, excluding areas with unsuitable soil types.

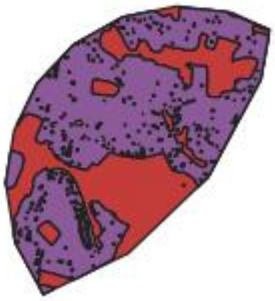
2. **Buffer Analysis**

- A **1 km buffer** was created around **Ananda Primary School** and a **2 km buffer** around **Sri Gunananda Vidyalaya** using the **Buffer Tool** in QGIS. This identified a potential suitability zone for the new R&D center.



3. **Clipping and Intersection**

- The intersection of the two buffers was clipped to identify the overlap, forming the base area for further suitability analysis.
- Land use data was also clipped to exclude areas under **tea, rubber**, or **coconut** cultivation.



4. **Building Footprint Analysis**
   - Existing buildings within the suitability zone were extracted using the **Intersect** tool to determine:
     - Total **number of buildings** (17 buildings).
     - Total **land area occupied** by buildings (~0.3 km²).
     - **Available land** for construction (~4.2 km²).
5. **Final Suitability Map**
   - The final **suitability map** was produced with all standard map elements (north arrow, scale bar, legend, title). The map was classified by land availability.

Results

The suitability zone identified for the R&D center was located in a semi-urban area within proximity to both schools. Key findings include:

- **Geo-Spatially Identified Area**: The zone covers **4.5 km²**, after excluding buildings and unsuitable land.
- **Total Buildings**: 17 existing buildings, primarily small residential structures.
- **Land Occupied by Buildings**: Approximately **0.3 km²** occupied by the existing buildings.
- **Available Land**: After subtracting building footprints, **4.2 km²** remains available for the R&D center.

The selected area meets all conditions: it is sufficiently distanced from educational institutions and avoids areas under agricultural use.

Critical Discussion

1. **Accessibility and Proximity**
   The area's proximity to both **Ananda Primary School** and **Sri Gunananda Vidyalaya** is beneficial for future educational collaborations. The **buffer zones** ensure that the research center remains accessible to students and educators while minimizing disruptions to school operations. Furthermore, proximity to urban infrastructure enhances the accessibility of the R&D center (World Bank, 2021).

2. **Land Use and Environmental Suitability**
   Excluding areas with **traditional export crops** ensures that the center does not interfere with Sri Lanka's agricultural economy. Soil analysis confirms that the land is suitable for development, with loamy soils and low flood risk, which are critical for construction (FAO, 2020).

3. **Urbanization and Infrastructure**
   The **semi-urban** nature of the site offers opportunities for easy access to utilities and transport networks. However, the presence of existing buildings presents challenges, such as the need for partial relocation or adaptive reuse of structures. These factors could affect construction timelines and budgets.

4. **Long-Term Sustainability**
   The location offers substantial space for future expansion of the research center. Additionally, its position within an urban area provides access to skilled labor and international research collaborations, facilitating growth and innovation. However, urbanization pressures may eventually limit future growth, suggesting the need for additional research hubs in rural or less developed areas (UNESCO, 2021).

Conclusion

The geospatial suitability analysis for the Sri Lanka R&D Center for Space Sciences successfully identified a **feasible location** within **4.2 km²** of available land. The analysis confirmed that the site meets accessibility, environmental, and land-use requirements, supporting both **strategic urban placement** and **sustainable development**. However, careful consideration of urban density and future expansion is recommended to ensure long-term functionality. The project aligns with Sri Lanka's goals of **becoming a regional leader in space sciences** and contributes significantly to the country's **knowledge-based economy**.

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Appendix A – Supporting Evidence for Task (a)

This appendix contains all the supporting outputs, tables, figures, and scripts generated from the statistical analysis of the University Quality Ranking (UQR.csv) dataset using R. Each file name corresponds to the saved outputs provided with the submission. The appendix is structured to align with the evidence requirements of Task (a) in the assessment brief.

Tables

- Table A1. Descriptive statistics and missing values (*01_descriptives.txt*)
- Table A2. Shapiro–Wilk normality test results (*02_normality_shapiro.txt*)
- Table A3. Pearson correlation matrix (*03_correlation_matrix.xlsx*)
- Table A4. Simple linear regression summaries (*05_simple_regressions.txt*)

- Table A5. Multiple regression summary and confidence intervals. (*06_multiple_regression_summary.txt*)
- Table A6. Variance Inflation Factors (VIF) (*08_vif.xlsx*)
- Table A7. Breusch–Pagan test results (*09_heteroscedasticity_bptest.txt*)
- Table A8. Welch two-sample t-test results by Institution Type (*11_ttest_URS_by_InstitutionType.txt*)
- Table A9. R session information (*99_sessionInfo.txt*)

Figures

- Figure A1. Correlation heatmap (03_correlation_heatmap.png)
- Figure A2. Scatterplot: Graduation Rate vs URS (04_scatter_Graduation.Rate...._vs_URS.png)
- Figure A3. Scatterplot: Employment Rate vs URS (04_scatter_Employment.Rate...._vs_URS.png)
- Figure A4. Scatterplot: Research Funding vs URS (04_scatter_Research.Funding..Million.USD.._vs_URS.png)
- Figure A5. Scatterplot: Faculty Salary vs URS (04_scatter_Faculty.Salary..Avg..._vs_URS.png)
- Figure A6. Scatterplot: Student–Faculty Ratio vs URS (04_scatter_Student.Faculty.Ratio_vs_URS.png)
- Figure A7. Scatterplot: Tuition Fees vs URS (04_scatter_Tuition.Fees..USD.._vs_URS.png)
- Figure A8. Regression diagnostics – Residuals vs Fitted (07_diagnostics_1_residuals_vs_fitted.png)
- Figure A9. Regression diagnostics – Normal Q-Q plot (07_diagnostics_2_qq.png)
- Figure A10. Regression diagnostics – Scale–Location plot (07_diagnostics_3_scale_location.png)
- Figure A11. Regression diagnostics – Cook’s distance (07_diagnostics_4_cooks.png)
- Figure A12. Boxplot of URS by Institution Type (10_boxplot_URS_by_InstitutionType.png)

Scripts

- Task a.R

Appendix B – Outputs for Task (b)

This appendix provides the supporting visual outputs generated during the geospatial analysis of Sri Lanka's school distribution using QGIS.

B1. Dataset Preparation

- **File Used:** XXX.csv (*Administrative Structure by District, 2022*)
- Screenshot: *B1_dataset_import.png*
- Description: CSV file containing district-wise counts of National Schools, Provincial Schools, and Total Schools imported into QGIS.

B2. Attribute Join

- **File Used:** task b.qgz (*QGIS Project File*)
- Screenshot: *B2_attribute_join.png*
- Description: District shapefile joined with CSV table using the common *District Name* field.

B3. Classification of Districts

- **File Generated:** Graduated color map (Total Schools)
- Screenshot: *B3_classification.png*
- Description: District polygons symbolized by the number of total schools using a graduated color scheme.

B4. Map Layout Design

- **Elements Included:** Title, Legend, Scale Bar, North Arrow, Labels
- Screenshot: *B4_map_layout.png*
- Description: Final QGIS map layout showing spatial distribution of National and Provincial schools by district.

B5. Final Map Output

- **File Saved:** *Sri_Lanka_School_Distribution_Map.png*
- Screenshot: *B5_final_map.png*
- Description: Exported high-resolution map used in the report, visualizing the geographic disparities in school distribution.

B6. Supporting Files

- **QGIS Project File:** task b.qgz
- **Dataset:** XXX.csv
- **Final Map Image:** Sri_Lanka_School_Distribution_Map.png

Appendix C – Outputs for Task (c)

This appendix presents the step-by-step outputs generated during the digitization and geo-referencing of the Ministry of Higher Education (MOHE) premises using QGIS.

C1. Raster Import

- **File Used:** Minstry of Higher Education_modified.tif
- Screenshot: *C1_raster_import.png*
- **Description:** The raster image of the MOHE site imported into QGIS as the base for geo-referencing.

C2. Geo-Referencing

- **Coordinate System:** EPSG:5234 (Sri Lanka Kandawala Grid)
- Screenshot: *C2_georeferencing.png*
- **Description:** Control points added and raster aligned to ensure spatial accuracy with OpenStreetMap basemap.

C3. Digitization of Features

- **Layers Created:**
  - *MOHE_Building (Polygon)*
  - *Service_Roads (Polyline)*
  - *Parking_Areas (Polygon)*
  - *Boundary (Polygon)*
- Screenshot: *C3_digitization.png*
- **Description:** Vector layers created through manual digitization of MOHE premises and surrounding infrastructure.

C4. Attribute Table

- **Fields Added:** *id, name, type, size*

- Screenshot: *C4_attribute_table.png*
- **Description:** Attribute data recorded for each digitized feature to enable querying and analysis.

C5. Map Layout Design

- **Elements Included:** Title, Legend, North Arrow, Scale Bar, Labels, Caption
- Screenshot: *C5_map_layout.png*
- **Description:** Final QGIS map layout showing the geo-referenced MOHE site with all relevant features.

C6. Final Map Export

- **File Saved:** *MOHE_Georeferenced_Map.png*
- Screenshot: *C6_final_map.png*
- **Description:** High-resolution map exported for inclusion in the main Task (c) report.

Appendix D – Outputs for Task (d)

This appendix provides the necessary supporting information and screenshots for Task (d), including the PostgreSQL database schema, SQL code, KML file integration, and the final QGIS map outputs.

D1. PostgreSQL Database Schema

- **Database Name**: SL_Uni_2025
- **Table**: mohe.university
- **Fields**:
    - id (Primary Key)
    - name (University name, string)
    - district (District name, string)
    - latitude (Latitude, numeric)
    - longitude (Longitude, numeric)
    - geom (Geospatial geometry, Point)

The **KML data** was loaded into PostgreSQL with a custom ogr2ogr tool and SQL COPY commands, creating a geom field as a point geometry for each university.

D2. KML File Integration

- **File Used**: university.kmz
- **Description**: The KML file contains the geographic coordinates (latitude and longitude) of Sri Lanka's 18 universities. The KML was converted into CSV format, and the data was then imported into PostgreSQL with spatial data. The **Point geometry** for each university was created using ST_SetSRID(ST_MakePoint(longitude, latitude), 4326).

KML Import Code Example:

ogr2ogr -f "PostgreSQL" PG:"host=localhost user=postgres dbname=SL_Uni_2025" university.kmz -nln mohe.university

This command reads the KML file and populates the university table with all relevant data and geographical coordinates.

D3. SQL Query for Data Retrieval

- **SQL Code to Retrieve Data**:

SELECT name, district, ST_AsText(geom) FROM mohe.university;

This query retrieves the names, districts, and spatial data (as text) for each university. It was used to verify the integration of the KML data into PostgreSQL.

D4. QGIS Map Screenshots

- **File Saved**: Sri_Lanka_University_Distribution_Map.png
- **Screenshots**:
  - *D4a_final_map.png*: Final QGIS map showing university distribution across districts, with labels and color symbology.
  - *D4b_buffer_analysis.png*: Screenshot of the 1 km buffer around Ananda Primary School and 2 km buffer around Sri Gunananda Vidyalaya.
  - *D4c_intersection_results.png*: Screenshot of the intersection analysis between buffer zones and land use filtering.

D5. Map Layout Design

The map was designed to include:

- **Title**: “Sri Lanka University Distribution by MOHE”
- **Legend**: Distinguishing universities by district

- **Scale Bar**: Representing geographic distances
- **North Arrow**: Indicating direction
- **Labels**: University names and district names

The map was saved in **PNG format** and included in the Task (d) report.

D6. Supporting Files

- **PostgreSQL Database**: SL_Uni_2025 (PostGIS extension enabled)
- **KML File**: university.kmz
- **QGIS Project File**: task d.qgz
- **Final Map Image**: Sri_Lanka_University_Distribution_Map.png

Appendix E – Outputs for Task (e)

This appendix contains the screenshots and descriptions of the workflow followed in **QGIS** and **PostGIS** for geospatial suitability analysis of the Sri Lanka R&D Center for Space Sciences.

E1. Data Import and Preparation

- **File Used**: task e.qgz (QGIS project file)
- **Description**: Import of the geospatial data layers (KML, land use, soil suitability) into QGIS.
- **Screenshot**: *E1_data_import.png*

E2. Buffer Analysis

- **Description**: A 1 km buffer around **Ananda Primary School** and 2 km buffer around **Sri Gunananda Vidyalaya** were created to define the proximity limits for the R&D center.
- **Screenshot**: *E2_buffer_analysis.png*

E3. Intersection and Clipping

- **Description**: Intersection of the buffer zones with land use and soil suitability layers. Areas classified as export crop lands were excluded from the final suitability zone.
- **Screenshot**: *E3_intersection_clipping.png*

E4. Building Footprint Analysis

- **Description**: Analysis of existing building footprints within the suitability zone to determine the land area already occupied by buildings and the available land for construction.
- **Screenshot**: *E4_building_analysis.png*

E5. Final Suitability Map

- **Description**: Final map showing the **suitable land for the R&D center** in relation to existing schools, land use, and building occupancy. This map was created with standard map elements (title, legend, scale bar, north arrow).
- **File Saved**: *Sri_Lanka_RnD_Center_Suitability_Map.png*
- **Screenshot**: *E5_final_suitability_map.png*

E6. Supporting Files

- **QGIS Project File**: task e.qgz
- **Final Suitability Map Image**: Sri_Lanka_RnD_Center_Suitability_Map.png
- **Land Use and Buffer Layers**: Provided with the project

This appendix provides the necessary supporting files for the suitability analysis and geospatial decision-making process. Each step (E1–E6) corresponds to the visual evidence of the analysis performed in QGIS.